High Performance Scientific Computing for Aerospace Project

11th November 2025 update

November 11th, 2025

Slide 1

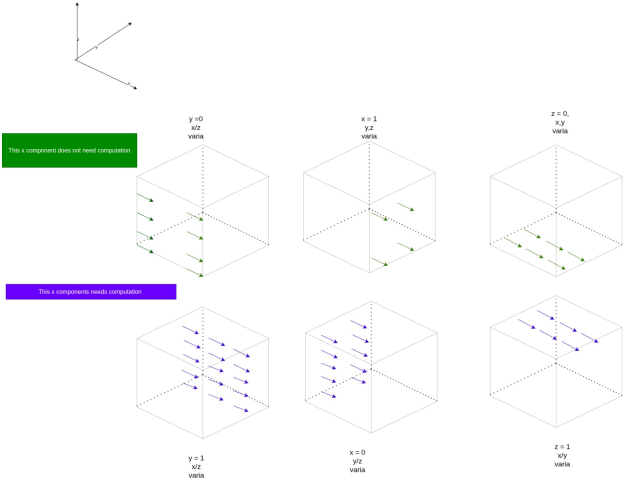
Content for slide 1 goes here.

Input File

Structure of the input file:

- **Mesh points:** Number of points in X, Y, Z directions
  - Example: `50 50 50` $\Rightarrow Nx = 50, Ny = 50, Nz = 50$
- **Time parameters:**
  - `dt` - Time step size, e.g., `0.01`
  - `T` - Total simulation time, e.g., `10.0`
- **Spatial step sizes:**
  - `dx dy dz` - Grid spacing in X, Y, Z, e.g., `0.5 0.5 0.5`
- **Initial conditions files:**
  - `u0_init_file` - File with initial velocity values
  - `p0_init_file` - File with initial pressure values
  - `k_file` - File with permeability values

Boundary conditions analysis



For each of the three steps of the algorithm for momentum equation, the Dirichlet Boundary Conditions are applied in the following way:

when solving for the component parallel to the decomposition direction:

$$\begin{bmatrix} 1 & 0 & 0 & \cdots & 0 \\ a & b & c & \ddots & \vdots \\ 0 & a & \ddots & \ddots & 0 \\ \vdots & \ddots & \ddots & b & c \\ 0 & \cdots & 0 & 0 & 1 \end{bmatrix} \begin{bmatrix} \eta_{1,x} \\ \eta_{2,x} \\ \vdots \\ \eta_{N-1,x} \\ \eta_{N,x} \end{bmatrix} = \begin{bmatrix} u_{0,j,k} + \left.\frac{\partial u}{\partial x}\right|_{(0,j,k)} \frac{\Delta x}{2} \\ r_{2,x} \\ \vdots \\ r_{N-1,x} \\ \text{BC value} \end{bmatrix}$$

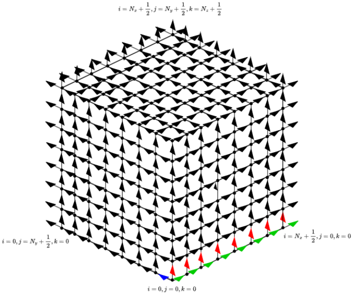
when solving for a component perpendicular to the decomposition direction:

$$\begin{bmatrix} 1 & 0 & 0 & \cdots & 0 \\ a & b & c & \ddots & \vdots \\ 0 & a & \ddots & \ddots & 0 \\ \vdots & \ddots & a & b & c \\ 0 & \cdots & 0 & a & b-c \end{bmatrix} \begin{bmatrix} \eta_{1,y} \\ \eta_{2,y} \\ \vdots \\ \eta_{N-1,y} \\ \eta_{N,y} \end{bmatrix} = \begin{bmatrix} \text{BC value} \\ r_{2,y} \\ \vdots \\ r_{N-1,y} \\ r_{N,y} - 2cu_{ex} \end{bmatrix}$$

Boundary Conditions: Indexing Pattern

We found a structural pattern in our staggered grid that simplifies boundary condition (BC) logic. Let x_0 be the solving direction and x_1 and x_2, with indices i_1 and i_2, the first and second direction along which the solver iterates. Then:

- **Normal Component:** i_1 is **even** AND i_2 is **even**.
- **Tangential Component:** i_1 is **odd** OR i_2 is **odd**.



This pattern holds for all **three velocity components** and allows for a **systematic implementation** of BCs. The logic is implemented in the code simply by modifying the **stride computation** for each direction, a task easily programmed with templates.

Future workflow

Code Validation

- Compute error
- Compute residual
- Convergence study

Parallelization Strategy

- Vectorization of the sequential code
- Parallelization of the code

Slide 5

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Slide 6

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